

Monitoring and Understanding Notebook-Based Learning with Jupyter - Appendix

A Dashboard for Student Self-Assessment - Appendix

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A QUESTIONNAIRE ITEMS

This appendix provides the exact wording of the items used in the evaluation survey to measure system usability, technology acceptance, and perceived cognitive workload.

A.1 System Usability Scale (SUS)

Participants rated the 10 items of Table A.1 on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

A.2 Technology Acceptance Model (TAM)

Participants rated the items of Table A.2 on a 7-point Likert scale (Extremely Unlikely to Extremely Likely). The construct was divided into Perceived Usefulness (PU) and Perceived Ease of Use (PEOU).

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Table A.1. Items of the System Usability Scale (SUS)

Item	Statement
SUS01	I think I would like to use this Dashboard frequently.
SUS02	I found this Jupyter Dashboard unnecessarily complex.
SUS03	I thought the Jupyter Dashboard was easy to use.
SUS04	I think that I would need the support of a technical person to use the Jupyter Dashboard.
SUS05	I found the various functions in this dashboard were well integrated.
SUS06	I thought there was too much inconsistency in this dashboard.
SUS07	I would imagine that most people would learn to use this dashboard very quickly.
SUS08	I found this dashboard very awkward to use.
SUS09	I felt very confident using this Jupyter Dashboard.
SUS10	I needed to learn a lot of things before I could get going with this dashboard.

Table A.2. Items of the Technology Acceptance Model (TAM)

Construct	Statement
PU	<i>Using the Jupyter Dashboard to assist my learning progress would...</i>
PU01	...enable me to accomplish tasks more frequently.
PU02	...improve my learning performance.
PU03	...increase my productivity.
PU04	...enhance my effectiveness during the task.
PU05	...make it easier to do my tasks.
PU06	...be useful.
PEOU	<i>How much do you agree with each statement?</i>
PEOU01	Learning to operate this Dashboard would be easy for me.
PEOU02	I would find it easy to get this Dashboard to do what I want it to do.
PEOU03	My interaction with the Dashboard would be clear and understandable.
PEOU04	I would find this Dashboard would be clear and understandable.
PEOU05	It would be easy for me to become skillful at using this Dashboard.
PEOU06	I would find this Dashboard easy to use.

A.3 NASA Task Load Index (NASA-TLX)

Participants rated the items of Table A.3 on a continuous scale from 0 to 20 to measure their perceived workload during the task.

Table A.3. Items of the NASA-TLX

Dimension	Question
Mental Demand	How mentally demanding was the task?
Physical Demand	How physically demanding was the task?
Temporal Demand	How hurried or rushed was the pace of the task?
Performance	How successful were you in accomplishing what you were asked to do?
Effort	How hard did you have to work to accomplish your level of performance?
Frustration	How insecure, discouraged, irritated, stressed, and annoyed were you?